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Introduction

This quick-start guide covers:

- How to start the calibration process
- How to calibrate cameras with the active wand (*IR version 2*)
- Effective wand-waving technique
- Calibration feedback and camera health (*in brief*)
- How to confirm a successful calibration

Important - this guide assumes you have already set your camera masks.
If you have not yet masked your cameras, please complete this through the steps in the masking guide.

This is the second of the three system preparation guides:

1. Masking your Cameras
2. ***Calibrating your Cameras***

3. Setting Volume Origin and Plane

Camera Calibration (Quick-Start Guide)

In order for you to begin capturing your subject's 3D motion tracking data, the cameras need to be calibrated to the environment you are working in.

The cameras are very sensitive to changes in temperature and light, as well as being physically disturbed. As such it is recommended that you calibrate at the beginning of any new day or capture session, and re-calibrate if the environment has changed or been disturbed.

Calibration is what allows each camera to determine the position of a tracked marker relative to itself, and the position of other cameras relative to itself. This positional data is synchronised between every camera in order to determine the 3D space (*the volume*) and where within that volume that the marker currently is.

Important: The recommended practice is perform a calibration at the start of each day, or when beginning a capture session if the environment has changed. Before calibrating your cameras you should always re-apply their masks, following the camera masking guide.

Calibration Prepartion

System State and Panel View

In order to calibrate your cameras you must first ensure that you are in the correct system state.

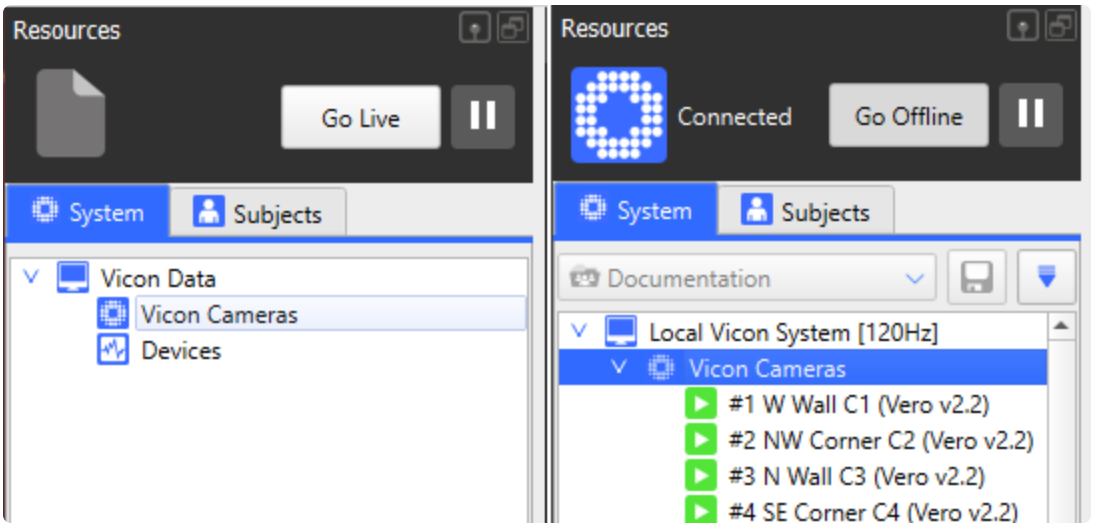
Offline mode supports:

1. subject preparation
2. labelling and editing
3. trial pipelines

Live mode supports:

1. system preparation
2. subject capture
3. trial capture

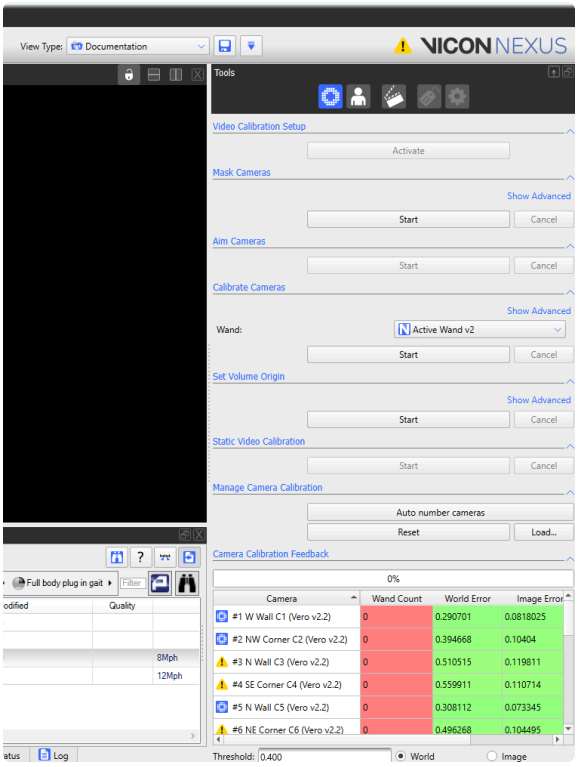
For calibration and system preparation purposes you must ensure the system is in **live mode**.



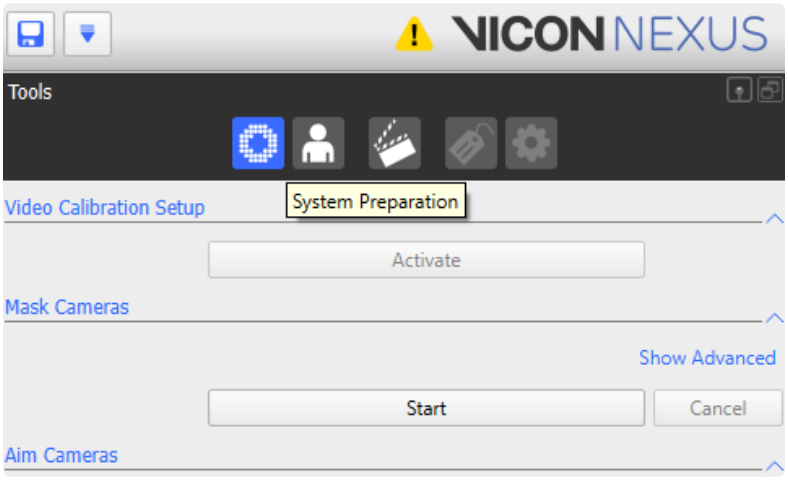
Vicon camera display state based on offline (left) and live (right) mode.

If your cameras are not currently displaying under the '**Vicon Cameras**' device group, click the '***Go live***' button to change system state.

Once you are ready to calibrate your cameras, look to **tools** panel on (by default) the right hand side, and make sure the **system preparation** tab is selected.



Tools panel, in system preparation view



How to select the system preparation tab from the tools panel

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Resetting Calibration

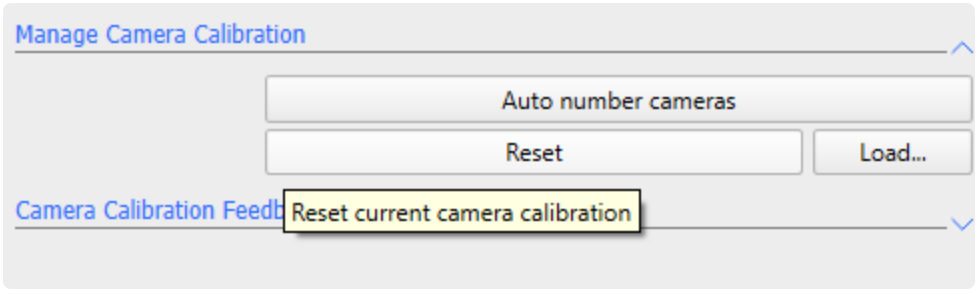
Calibrations can be repeated to improve accuracy.

If you attempt a calibration at the start of a new session without resetting the existing calibration data, you will instead influence the previous calibration.

Capture environments change over time and what was accurate earlier may no longer be accurate. In order to ensure previous calibrations do not influence the day's calibration, you will need to remove the calibration profile from every camera.

If this is not a brand new calibration, do not reset camera calibration. This will remove all previous camera information about their relative position to each other and the volume.

At the bottom of the **system preparation** tab you will see the **manage camera calibration** section.



Camera calibration behaviour should be managed with care

Do not select 'auto number cameras' - this will reassign camera IDs and put them out of sync with the lab's physical labelling.

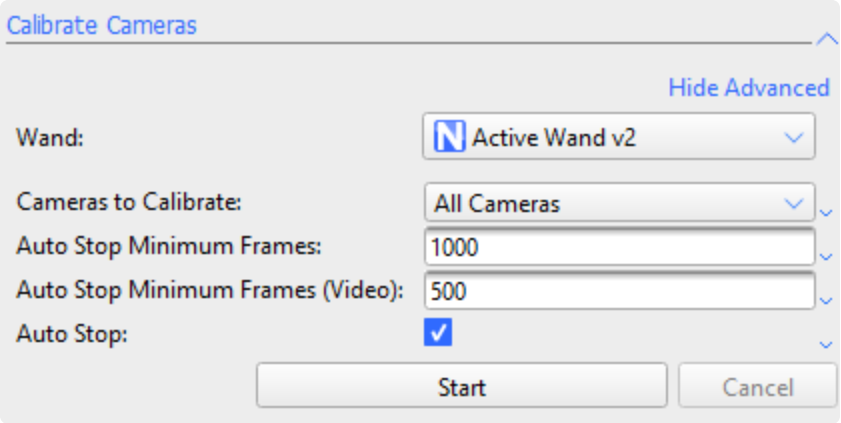
Click the **Reset** button in . If you are in 3D Perspective view, and the cameras were previously calibrated, you will see the cameras either disappear or reposition into a straight line

(depending on view option selected).

This represents that the cameras no longer know where each other are, relative to themselves.

Calibration Parameters

Underneath the '**Calibrate Cameras**' section you will see "*Show/Hide Advanced*" - ensure this is toggled to display all calibration properties.



Calibrate cameras section with all properties displayed

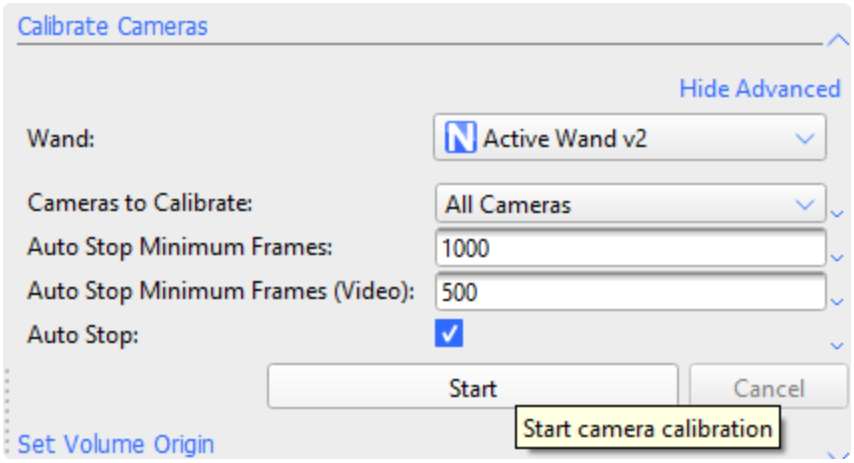
The properties here are as follows:

- **Cameras to Calibrate**: *whether to calibrate all or specific cameras.*
- **Auto Stop Minimum Frames**: *the minimum number of frames a camera has to capture to end calibration.*
- **Auto Stop Minimum Frames (Video)**: *as above, but for video cameras - not relevant to this guide.*
- **Auto Stop** (*toggle*): *whether to use the auto-stop functionality above.*

If this is your first calibration, use the following presets:

- **Cameras to Calibrate**: All Cameras
- **Auto Stop Minimum Frames**: 1000
- **Auto Stop**: Yes ☑

Once you are happy with these settings, start your camera calibration by clicking the *Start* button.

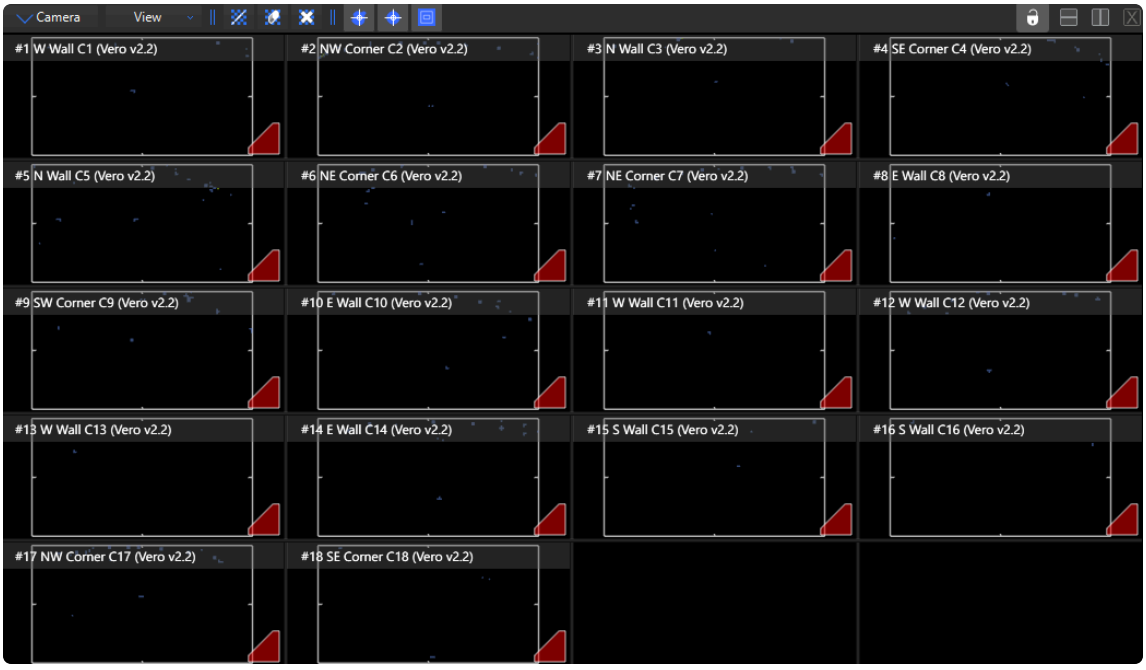


Position of the calibration start button

The Calibration Process

Once you begin your your calibration the Nexus window will automatically switch to display the camera view for every camera registered.

This includes disabled, disconnected, and missing cameras. When calibrating with '**all cameras**' ensure the cameras present in the System/Resources view are the cameras you wish to calibrate.



View of all cameras ready to begin calibration

The Active Wand

To calibrate your cameras you must wave the **active wand** within the capture volume (see below).

Ensure the active wand is in '**strobe**' not '**continuous**' mode before you begin wand waving. One switch controls the wand LED mode, and one switch controls whether the wand is **on** or **off**.



Image of active wand IRv2

The Capture Volume

The capture volume is the area that is covered by the camera setup, a guideline to which the cameras are arranged to have the maximum possible view of. When calibrating cameras, or capturing trials, you should remain within this volume.

You can use the **Options shortcut (F7 by default)** to access the Nexus options menu, and toggle the volume display **on** or **off**.

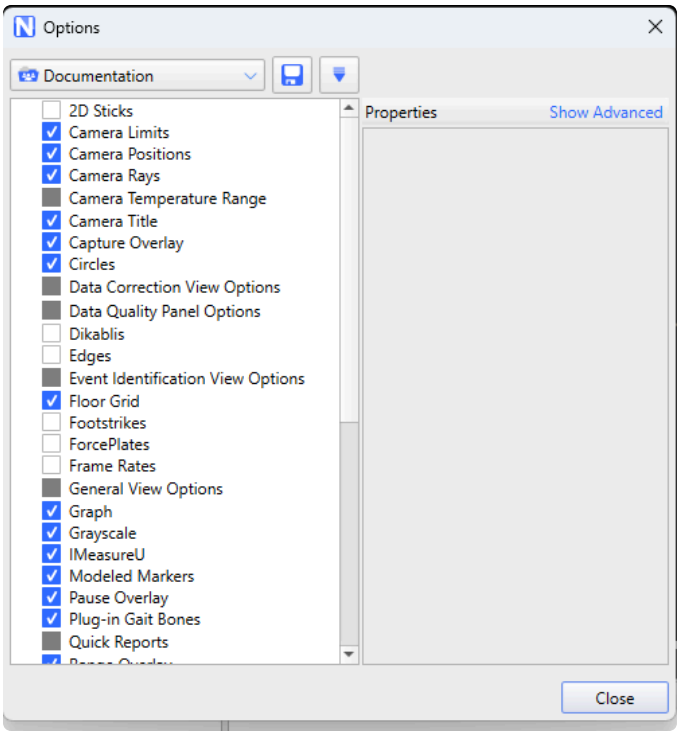
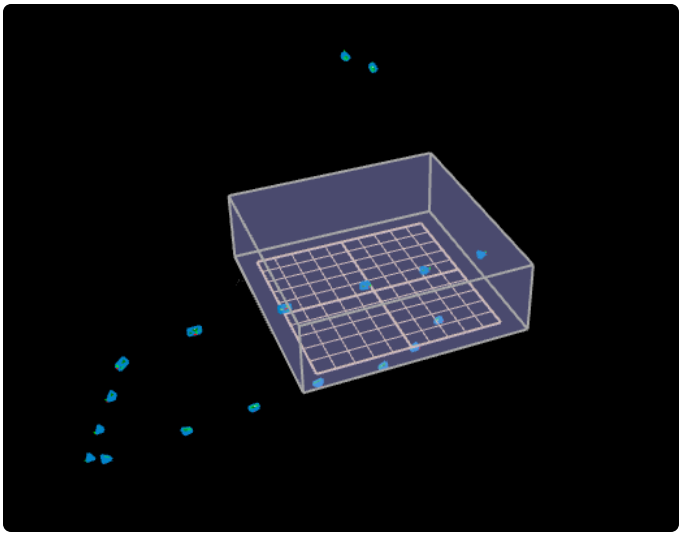


Image of the Vicon Nexus 2.18 options menu

This display is prominent on the **view panel**, and many other functions within the Vicon Nexus software are simpler to work with whilst the display is toggled off.

Whilst in the **3d perspective view**, the target volume will appear as a box around the floor grid.


Image of the target volume in Vicon Nexus 2.18.

INB0104 only: Within the INB0104 laboratory the target volume is mapped to the edges of the square denoted by white plastic tape on the north/south edges, and yellow/black striped warning tape on the east/west edges. Calibration, and capture, should take place within these boundaries.

Calibration Frame Feedback

On starting calibration with a specified auto-stop frame count, all the cameras within your system will change their lights based on the amount of calibration frames they have received.

Initially the cameras will not show any lights, indicating that they have not received any calibration frames.

Once you begin calibration the camera lights within the room will begin to flash blue, indicating they are observing calibration frames.



Camera during calibration that has received some calibration frames but needs more.



Camera during calibration that has received almost all calibration frames it needs.

As the cameras accept calibration frames they will begin to flash quicker, until switching to a solid green hue once they have reached their minimum frame value for the auto-stop setting.

Using the defaults given in the previous part of this guide (**auto-stop: 1000**), this would be after they have tracked 1000 frames of the wand.



This camera has reached their target calibration frames.

When all cameras have met the auto-stop calibration frame goal, they will all turn a solid blue colour.

Wand Waving

To calibrate the cameras you must wave the wand within the volume, so that they are able to see the wand LEDs, until all cameras receive enough frames to reach their auto-stop minimum. This section covers the minimum you will need to know to calibrate your system; the dedicated wand waving guide covers this topic in more detail.

To ensure a successful calibration:

- Start from the inside of the volume, and 'paint' the space.
- Move outwards from the inside and attempt to cover
- Hold the wand away from your body, as to avoid occlusion from your body.
- Turn your body so that different cameras are able to see the wand LEDs.
- Wave the wand held flat, with the LEDs pointing upward, facing towards the camera.
- Move the wand in a circular motion, ensuring that you move the wand up and down to cover the volume.

Your goal whilst calibrating is to provide as much visual data to the cameras as possible, covering the space and ensuring as many cameras as possible see the wand within every area of the volume.

Do not spend disproportionate time on one area of the volume until the volume has been fully covered.

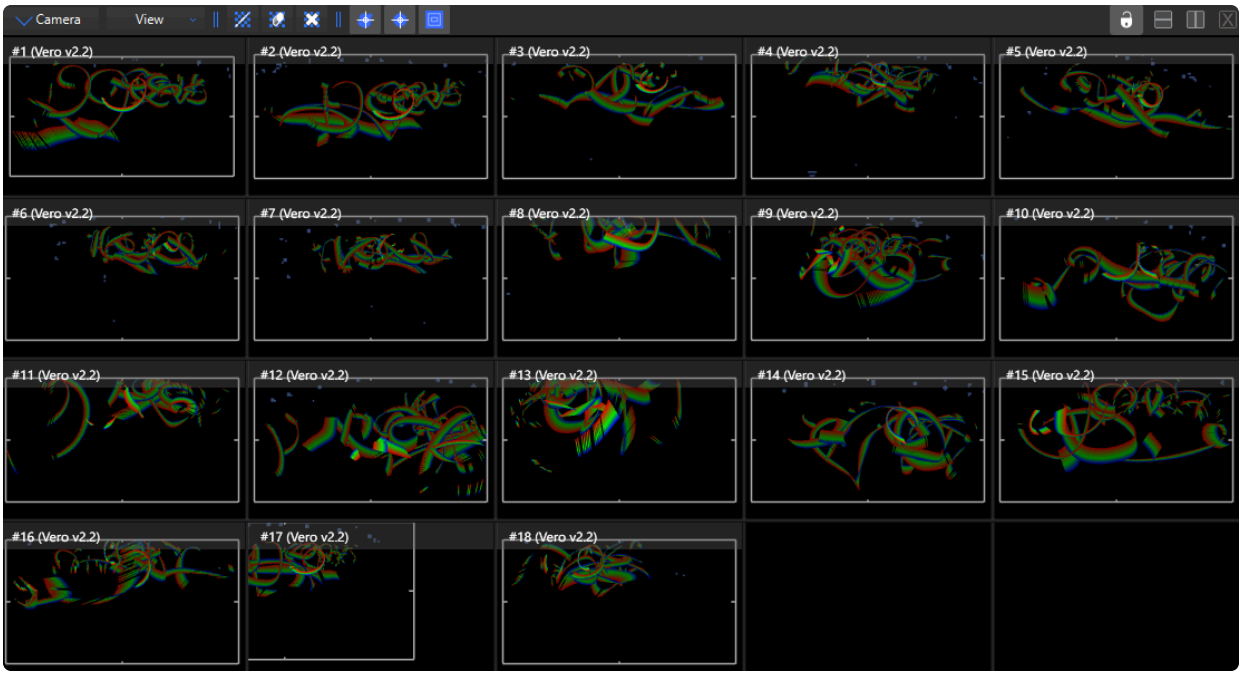
Calibration occurs on a per-camera basis, and any camera that is not effectively calibrated will provide misleading information for other cameras.

Further information: 3D captures of wand waving technique are available with the documentation dataset.

Successful Calibration

Once you have completed calibration, all camera lights will change from green to blue, and Nexus will begin the calibration feedback process.

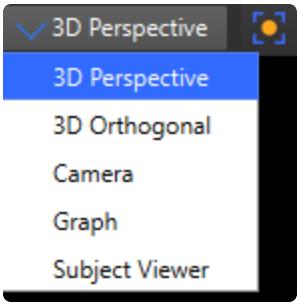
The Nexus window will continue to display the wand wave trail data until you leave this view. This is a visual representation of where the cameras saw the wand whilst you were calibrating.



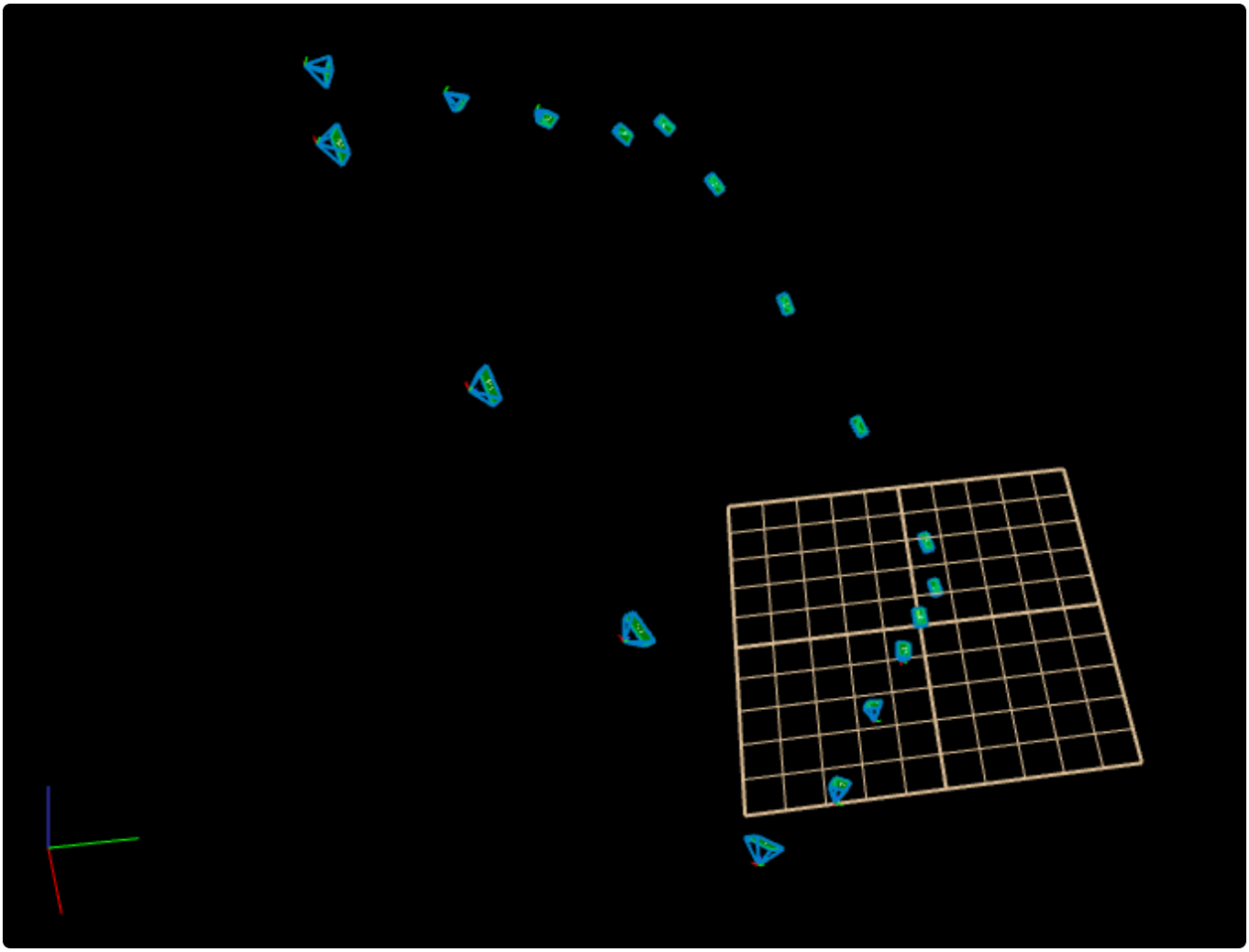
Wand wave data output

The above wand wave trail data is not saved by Vicon Nexus.
If you wish to preserve this image for any data collection purposes - take a screenshot of it now.

If you switch to '3d perspective' view you will be able to see immediate effects from the calibration.



Once you are in 3D perspective view, note the position of each camera in the software, relative to each other.

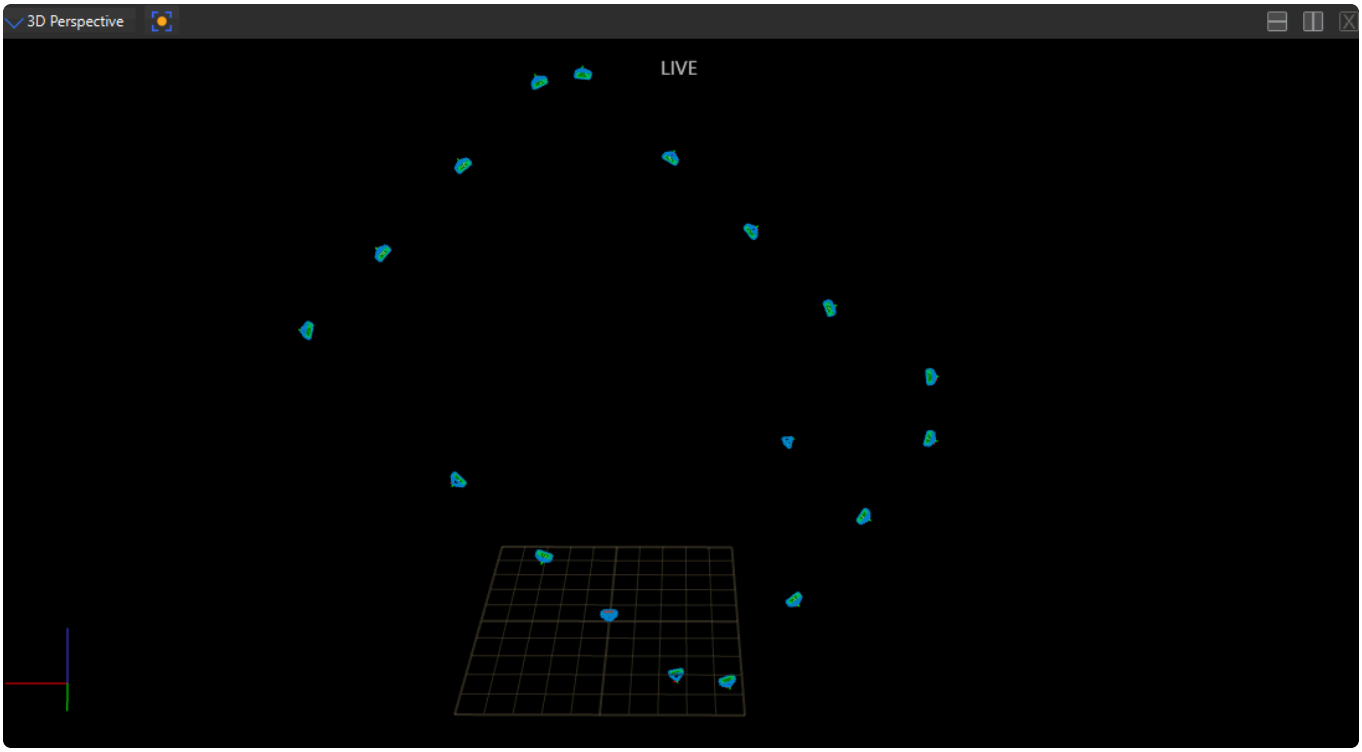


View of a successful calibration - all cameras are spaced as per setup in the room.

If this is representative of their position to each other, the calibration was successful for determining camera position.

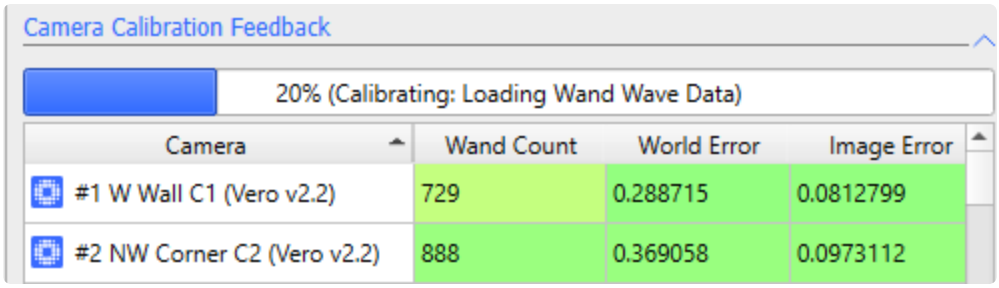
This is not a guarantee of successful calibration, but it is the minimum to expect at this stage.

If any camera is unexpectedly out of position, consider that the calibration needs to be reattempted.

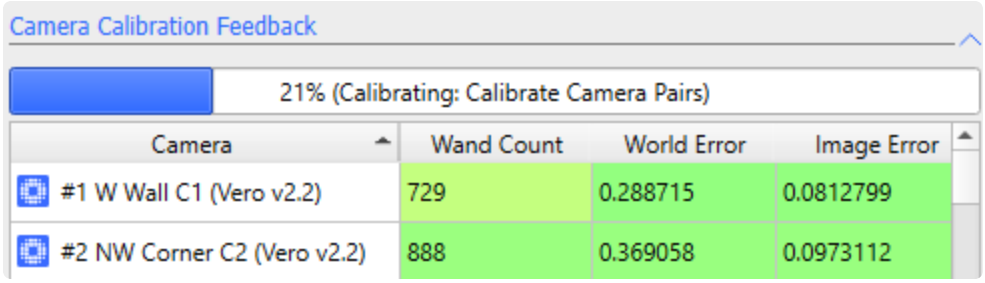


View of an unsuccessful calibration - one camera is isolated, away from where it should be within the camera ordering (left hand side).

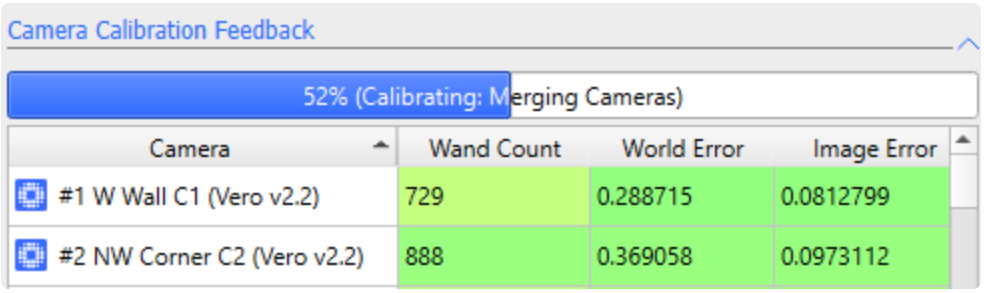
At the bottom of the **tools panel**, underneath the *camera calibration feedback* section, you will see a progress bar. This is Nexus determining the camera calibration from the wand wave data you have just provided.



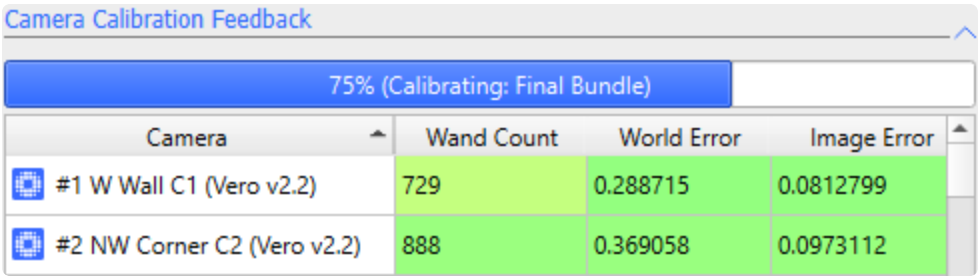
Calibration feedback initial state



Calibration feedback progressing



Calibration feedback merging information from cameras



Calibration feedback readying to output data

Once this has been completed you are ready to set the volume origin and floor plane.